Formulations of lot sizing problems

Outline

Overview of the family

ILP models for the Capacitated Lot Sizing Problem

A formulation with a big-M link

The Linear Programming relaxation of the big-M formulation

A shortest-path formulation

Comparing the formulations

ILP models for the variants

Typical mistakes

In summary

Learning objectives

At the end of this chapter you will be able to:

- write the big-M formulation of the capacitated lot sizing problem and compute a weak and a strong big-M value;
- write the shortest-path formulation and interpret its variables as the arcs of a network;
- represent a production plan as a combination of paths of the network;
- compare the LP bounds of the formulations on the reference instance;
- write the ILP model of the uncapacitated lot sizing problem;
- extend the big-M formulation to several products with setup times, to the initial stock, to overtime and to unsatisfied demand, and compute the strong big-M values of each model.

Prerequisites

- From the module *Integer linear programming*, the chapters «Integer linear programming» and «Primal and dual bounds»: ILP problems, LP relaxations and lower bounds.
- From this module, the chapter «Lot sizing problems»: the problem, its assumptions, the plans and the reference instance.

Overview of the family

Introduction: overview of the family

- This chapter writes the models of the lot sizing problems: the capacitated problem and its variants
- All the models share the same variables and the same kinds of constraints
- We first look at the family as a whole: its variables, its constraints and the list of its members

Variables, constraints and objective

- The big-M models of the lot sizing problems use, for every period $t \in T$, an integer **production variable** x_t, an integer **inventory variable** s_t and a binary **setup variable** y_t.
- The **flow balancing constraints** link the inventories of consecutive periods to the production and to the demand; the setup variables are linked to the production by big-M constraints.
- The objective function minimizes the total cost of setups, production and holding. The shortest-path formulation of the CLS replaces the production and inventory variables with the arcs of a network.

Flow models with setups

- The lot sizing models are **flow models over the periods**, with binary setup variables that switch the production on.
- The members of the family add products, setup times, an initial stock, overtime or unsatisfied demand to the same balance and big-M constraints.

The members of the family

The members of the family, with the names of their models and the sections that write them:

Problem	**Models**	**Formulation**
Section «A formulation with a big-M link»		
CLS	$\mathrm{ILP}_{\mathrm{CLSP}}$	big-M formulation
Section «A shortest-path formulation»		
CLS	$\mathrm{MIP}^{\mathrm{SP}}_{\mathrm{CLSP}}$	shortest-path formulation
Section «ILP models for the variants»		
ULS	$\mathrm{ILP}_{\mathrm{ULS}}$	big-M, no capacity
MLS	$\mathrm{ILP}^{P}_{\mathrm{CLSP}}$	big-M, setup times
LS-SI	$\mathrm{ILP}^{\mathrm{SI}}_{\mathrm{CLSP}}$	big-M, initial stock variables
LS-OT	$\mathrm{ILP}^{\mathrm{OT}}_{\mathrm{CLSP}}$	big-M, overtime variables
LS-UD	$\mathrm{ILP}^{\mathrm{UD}}_{\mathrm{CLSP}}$	big-M, unsatisfied demand variables

Summary

- Every big-M model of the family uses production variables x_t, inventory variables s_t and setup variables y_t
- The flow balancing constraints link inventories, production and demand; big-M constraints link the setups to the production
- The variants add products, setup times, an initial stock, overtime or unsatisfied demand to the same constraints

ILP models for the Capacitated Lot Sizing Problem

Introduction: ILP models for the CLS

- We now start from the capacitated lot sizing problem (CLS), the main member of the family
- It has two different formulations, and we will write both of them
- Two correct models of the same problem can give very different lower bounds: this is the thread of the next sections

The CLS and its two formulations

- The Capacitated Lot Sizing Problem and its reference instance are defined in the chapter «Lot sizing problems».
- The CLS has two formulations: the big-M formulation $\mathrm{ILP}_{\mathrm{CLSP}}$ (section «A formulation with a big-M link») and the shortest-path formulation $\mathrm{MIP}^{\mathrm{SP}}_{\mathrm{CLSP}}$ (section «A shortest-path formulation»).
- The section «Comparing the formulations» compares the bounds of their Linear Programming relaxations on the reference instance.

Summary

- The CLS has a big-M formulation $\mathrm{ILP}_{\mathrm{CLSP}}$ and a shortest-path formulation $\mathrm{MIP}^{\mathrm{SP}}_{\mathrm{CLSP}}$
- We will compare the bounds of their LP relaxations on the reference instance

A formulation with a big-M link

Introduction: a formulation with a big-M link

- We write the first ILP model of the CLS, with production, inventory and setup variables
- The setups are linked to the production by a big-M constraint
- The value of M matters: we compute a weak and a strong value
- We close with the model on the reference instance and its optimal solution

Total demands and variables

- Given a set $T = \{1, 2, \ldots, m\}$ of periods with demands d_t, for $j \leq t$ we write $D_{j,t} = \sum_{k=j}^{t} d_k$ for the total demand of the periods $j, j+1, \ldots, t$.
- We introduce the following **integer variables** and **binary variables**, for every period $t \in T$:

$$x_t \in \mathbb{Z}_{\geq 0} \text{ (production)},$$
$$s_t \in \mathbb{Z}_{\geq 0} \text{ (inventory at the end of the period)},$$

$$y_t = \begin{cases} 1 & \text{if there is a setup in period } t \\ 0 & \text{otherwise} \end{cases}$$

The inventory at the beginning of the horizon is the constant s_0.

The big-M constraint

- The setup variables are linked to the quantities by the logical implication

$$x_t > 0 \quad \Longrightarrow \quad y_t = 1,$$

  modeled by the **big-M constraint** $x_t \leq M_t\, y_t$, where M_t is an upper bound on the production of period t.

- If the setups are ignored, the best plan produces in every period exactly its demand: it has no inventory and, in the reference instance, cost 0.

- This is a very weak lower bound: the setup costs must be modeled.

The big-M formulation $\mathrm{ILP}_{\mathrm{CLSP}}$

Using these variables, an Integer Linear Programming (ILP) model for the capacitated lot sizing problem (CLS) reads as follows:

$$\min \quad \sum_{t \in T} (sc_t\, y_t + vc_t\, x_t + hc_t\, s_t)$$

$$\begin{aligned}
\text{subject to} \quad & s_{t-1} + x_t = d_t + s_t, && \forall t \in T, \\
& x_t \leq M_t\, y_t, && \forall t \in T, \\
& x_t \leq cap_t, && \forall t \in T, \\
& s_0 = s_m = 0, && \\
& x_t \in \mathbb{Z}_{\geq 0}, && \forall t \in T, \\
& s_t \in \mathbb{Z}_{\geq 0}, && \forall t \in T, \\
& y_t \in \{0, 1\}, && \forall t \in T.
\end{aligned}$$

Description of the model $\mathrm{ILP}_{\mathrm{CLSP}}$

- The linear objective function minimizes the cost of the setups, of the production and of the holding;
- constraints $s_{t-1} + x_t = d_t + s_t$ are the **flow balancing constraints**: the demand of period t is satisfied by the production of the period and by the inventory of the previous period, and the surplus becomes the inventory at the end of period t (m linear constraints);
- constraints $x_t \leq M_t\, y_t$ link the quantities with the setups, and constraints $x_t \leq cap_t$ are the capacity constraints (m linear constraints each);
- constraint $s_0 = s_m = 0$ imposes an empty inventory at the beginning and at the end of the horizon;
- finally, the domain constraints define the variables of the ILP model.

This ILP model is referred to as the $\mathrm{ILP}_{\mathrm{CLSP}}$ model, or **big-M formulation**.

Weak and strong big-M values

- **A weak value.** Since $s_m = 0$ and the inventories are non-negative, the production of period t cannot exceed the demand of the periods $t, \ldots, m$: $M_t^{\text{dem}} = D_{t,m}$ is a valid value.
- **A second value.** By the capacity constraints $x_t \leq cap_t$, $M_t^{\text{cap}} = cap_t$ is also valid.
- **A strong value.** The best value is the smallest of the two:

$$M_t = \min \{cap_t,\ D_{t,m}\}\,.$$

Big-M values and LP relaxation

- **The smaller the big-M value, the stronger the LP relaxation**: the constraint $x_t \leq M_t y_t$ cuts off the fractional values of y_t below x_t/M_t.
- A value that is too small, however, cuts off feasible plans.

Example: the reference instance (1/2)

We consider $m = 5$ periods, with the following data:

period t	1	2	3	4	5
demand d_t	2	1	3	4	1
capacity cap_t	4	4	4	4	4
setup cost sc_t	50	50	50	50	50
production cost vc_t	0	0	0	0	0
holding cost hc_t	10	10	10	10	10

This is the reference instance of the chapter «Lot sizing problems».

Example: the reference instance (2/2)

The big-M values are:

period t	1	2	3	4	5
$M_t^{\text{dem}} = D_{t,5}$	11	9	8	5	1
$M_t^{\text{cap}} = cap_t$	4	4	4	4	4
M_t	4	4	4	4	1

We refer to this CLS instance as the **reference instance**.

Example: the $\mathrm{ILP}_{\mathrm{CLSP}}$ model (1/3)

The $\mathrm{ILP}_{\mathrm{CLSP}}$ model with the strong big-M values for the CLS reference instance is ($s_0 = 0$ is a constant, and it does not appear):

$$
\begin{aligned}
\text{min} \quad & 50\,y_1 + 50\,y_2 + 50\,y_3 + 50\,y_4 + 50\,y_5 + 10\,s_1 + 10\,s_2 \\
& +10\,s_3 + 10\,s_4 + 10\,s_5 \\
\text{subject to} \quad & x_1 - s_1 = 2 \\
& s_1 + x_2 - s_2 = 1 \\
& s_2 + x_3 - s_3 = 3 \\
& s_3 + x_4 - s_4 = 4 \\
& s_4 + x_5 - s_5 = 1
\end{aligned}
$$

Example: the $\mathrm{ILP}_{\mathrm{CLSP}}$ model (2/3)

$$
\begin{aligned}
x_1 - 4\,y_1 &\leq 0 \\
x_2 - 4\,y_2 &\leq 0 \\
x_3 - 4\,y_3 &\leq 0 \\
x_4 - 4\,y_4 &\leq 0 \\
x_5 - y_5 &\leq 0 \\
x_1 &\leq 4 \\
x_2 &\leq 4 \\
x_3 &\leq 4 \\
x_4 &\leq 4 \\
x_5 &\leq 4
\end{aligned}
$$

Example: the $\mathrm{ILP}_{\mathrm{CLSP}}$ model (3/3)

$$
\begin{array}{lcl}
s_5 & = & 0 \\
x_1,\ x_2,\ x_3,\ x_4,\ x_5 & \in & \mathbb{Z}_{\geq 0} \\
s_1,\ s_2,\ s_3,\ s_4,\ s_5 & \in & \mathbb{Z}_{\geq 0} \\
y_1,\ y_2,\ y_3,\ y_4,\ y_5 & \in & \{0,1\}
\end{array}
$$

An optimal solution $(\tilde{\boldsymbol{y}}, \tilde{\boldsymbol{x}}, \tilde{\boldsymbol{s}})(\mathrm{ILP}_{\mathrm{CLSP}})$ is:

$$\tilde{\boldsymbol{y}} = (1,0,1,1,0), \qquad \tilde{\boldsymbol{x}} = (3,0,4,4,0), \qquad \tilde{\boldsymbol{s}} = (1,0,1,1,0)$$

- The optimal value $z(\mathrm{ILP}_{\mathrm{CLSP}})$ is $3 \cdot 50 + 3 \cdot 10 = 180$, the cost of the optimal plan found by enumeration in the chapter «Lot sizing problems».
- The weak big-M values give the same optimal solution.

Summary

The big-M formulation $\mathrm{ILP}_{\mathrm{CLSP}}$:

- Integer variables x_t, s_t and binary variables y_t; flow balancing constraints and capacity constraints
- The link $x_t \leq M_t\, y_t$ models the implication $x_t > 0 \Rightarrow y_t = 1$

The big-M values:

- $D_{t,m}$ and cap_t are both valid; the strong value is $M_t = \min\{cap_t, D_{t,m}\}$
- The smaller the value, the stronger the LP relaxation

The reference instance: optimal plan $(3, 0, 4, 4, 0)$, of cost 180

The Linear Programming relaxation of the big-M formulation

Introduction: the LP relaxation of the big-M formulation

- The LP relaxation of an ILP model gives a lower bound on its optimal value
- In the big-M formulation the bound depends on the values M_t
- We compute the bound of the three choices M_t^{dem}, M_t^{cap} and M_t on the reference instance

Relaxing the integrality

- To obtain a Linear Programming (LP) relaxation of the $\mathrm{ILP}_{\mathrm{CLSP}}$ model, we relax the integrality constraints of the variables into non-negativity constraints and the bounds $y_t \leq 1$.
- The obtained LP model is written in the next slide.

The LP relaxation of the big-M formulation

$$\min \quad \sum_{t \in T} (sc_t\, y_t + vc_t\, x_t + hc_t\, s_t)$$

$$\begin{aligned}
\text{subject to} \quad & s_{t-1} + x_t = d_t + s_t, && \forall t \in T, \\
& x_t \leq M_t\, y_t, && \forall t \in T, \\
& x_t \leq cap_t, && \forall t \in T, \\
& s_0 = s_m = 0, && \\
& y_t \leq 1, && \forall t \in T, \\
& x_t \geq 0, && \forall t \in T, \\
& s_t \geq 0, && \forall t \in T, \\
& y_t \geq 0, && \forall t \in T.
\end{aligned}$$

Three LP models

- This LP model is referred to as the $\mathrm{LP}^{\mathrm{dem}}_{\mathrm{CLSP}}$, $\mathrm{LP}^{\mathrm{cap}}_{\mathrm{CLSP}}$ or $\mathrm{LP}^{\mathrm{strong}}_{\mathrm{CLSP}}$ model, when the big-M values are M_t^{dem}, M_t^{cap} or the strong values M_t.
- The three models differ only in the coefficients of the constraints $x_t \leq M_t\, y_t$.
- We now solve the three of them on the reference instance.

Example: big-M values and LP bounds (1/4)

The LP model for the CLS reference instance is:

$$
\begin{aligned}
\text{min} \quad & 50\,y_1 + 50\,y_2 + 50\,y_3 + 50\,y_4 + 50\,y_5 + 10\,s_1 + 10\,s_2 \\
& +10\,s_3 + 10\,s_4 + 10\,s_5 \\
\text{subject to} \quad & x_1 - s_1 = 2 \\
& s_1 + x_2 - s_2 = 1 \\
& s_2 + x_3 - s_3 = 3 \\
& s_3 + x_4 - s_4 = 4 \\
& s_4 + x_5 - s_5 = 1
\end{aligned}
$$

Example: big-M values and LP bounds (2/4)

$$
\begin{aligned}
x_1 - M_1\, y_1 &\leq 0 \\
x_2 - M_2\, y_2 &\leq 0 \\
x_3 - M_3\, y_3 &\leq 0 \\
x_4 - M_4\, y_4 &\leq 0 \\
x_5 - M_5\, y_5 &\leq 0 \\
x_1 &\leq 4 \\
x_2 &\leq 4 \\
x_3 &\leq 4 \\
x_4 &\leq 4 \\
x_5 &\leq 4
\end{aligned}
$$

Example: big-M values and LP bounds (3/4)

$$
\begin{aligned}
s_5 &= 0 \\
y_1 &\leq 1 \\
y_2 &\leq 1 \\
y_3 &\leq 1 \\
y_4 &\leq 1 \\
y_5 &\leq 1 \\
x_1,\ x_2,\ x_3,\ x_4,\ x_5 &\geq 0 \\
s_1,\ s_2,\ s_3,\ s_4,\ s_5 &\geq 0 \\
y_1,\ y_2,\ y_3,\ y_4,\ y_5 &\geq 0
\end{aligned}
$$

with the values $M_1, \ldots, M_5$ of the table of the reference instance.

Example: big-M values and LP bounds (4/4)

The three choices give the following optimal solutions $(\tilde{\boldsymbol{y}}, \tilde{\boldsymbol{x}}, \tilde{\boldsymbol{s}})$, with $\tilde{s}_5 = 0$:

Model	$\tilde{\boldsymbol{y}}$	$\tilde{\boldsymbol{x}}$	$(\tilde{s}_1, \ldots, \tilde{s}_4)$	optimal value
$\mathrm{LP}^{\mathrm{dem}}_{\mathrm{CLSP}}$	$(\frac{2}{11}, \frac{1}{9}, \frac{1}{2}, \frac{4}{5}, 0)$	$(2, 1, 4, 4, 0)$	$(0, 0, 1, 1)$	$\frac{9865}{99} \approx 99.65$
$\mathrm{LP}^{\mathrm{cap}}_{\mathrm{CLSP}}$	$(\frac{1}{2}, \frac{1}{4}, \frac{3}{4}, 1, \frac{1}{4})$	$(2, 1, 3, 4, 1)$	$(0, 0, 0, 0)$	$\frac{275}{2} = 137.5$
$\mathrm{LP}^{\mathrm{strong}}_{\mathrm{CLSP}}$	$(\frac{1}{2}, \frac{1}{4}, 1, 1, 0)$	$(2, 1, 4, 4, 0)$	$(0, 0, 1, 1)$	$\frac{315}{2} = 157.5$

- In every solution $\tilde{y}_t = \tilde{x}_t / M_t$: the smaller M_t, the larger the cost of the setups paid by the LP.
- The integer optimal solution of the $\mathrm{ILP}_{\mathrm{CLSP}}$ model has value $z(\mathrm{ILP}_{\mathrm{CLSP}}) = 180$.

Summary

- The LP relaxation of $\mathrm{ILP}_{\mathrm{CLSP}}$ replaces the integrality of the variables with non-negativity and the bounds $y_t \leq 1$
- In every LP solution $\tilde{y}_t = \tilde{x}_t / M_t$: the setups are paid only in part
- On the reference instance: 99.65 with M_t^{dem}, 137.5 with M_t^{cap}, 157.5 with the strong values, against the optimal value 180

A shortest-path formulation

Introduction: a shortest-path formulation

- We write a second formulation of the CLS, which does not decide the quantities directly
- Its variables are the arcs of a network: a plan becomes a path, or a combination of paths
- The setups are linked to the arcs without any big-M value

The variables zv_{tk} and their costs

- The second formulation does not decide the quantities directly. For every $t, k \in T$, with $k \geq t$, we introduce the following **continuous variable**:

  $zv_{tk} \geq 0$, the fraction of the demand $D_{t,k}$ of the periods $t, \ldots, k$ that is produced in period t.

- Producing in period t the demand of the periods $t, \ldots, k$ costs the production of $D_{t,k}$ units and the holding of the units that wait for their period:

$$cv_{tk} = vc_t \; D_{t,k} + \sum_{s=t+1}^{k} \sum_{u=t}^{s-1} hc_u \, d_s.$$

The shortest-path formulation $\mathrm{MIP}^{\mathrm{SP}}_{\mathrm{CLSP}}$

Using these variables and the binary variables y_t, a Mixed Integer Linear Programming (MIP) model for the CLS reads as follows:

$$\min \quad \sum_{t \in T} sc_t \, y_t + \sum_{t \in T} \sum_{k=t}^{m} cv_{tk} \, zv_{tk}$$

$$\text{subject to} \quad \sum_{k=1}^{m} zv_{1k} = 1,$$

$$\sum_{u=1}^{t-1} zv_{u,t-1} - \sum_{k=t}^{m} zv_{tk} = 0, \qquad \forall t \in T \setminus \{1\},$$

$$\sum_{k=t}^{m} zv_{tk} \leq y_t, \qquad \forall t \in T,$$

$$\sum_{k=t}^{m} D_{t,k} \, zv_{tk} \leq cap_t, \qquad \forall t \in T,$$

The model $\mathrm{MIP}^{\mathrm{SP}}_{\mathrm{CLSP}}$: domains and description

$$zv_{tk} \geq 0, \qquad \forall t, k \in T,\ k \geq t,$$

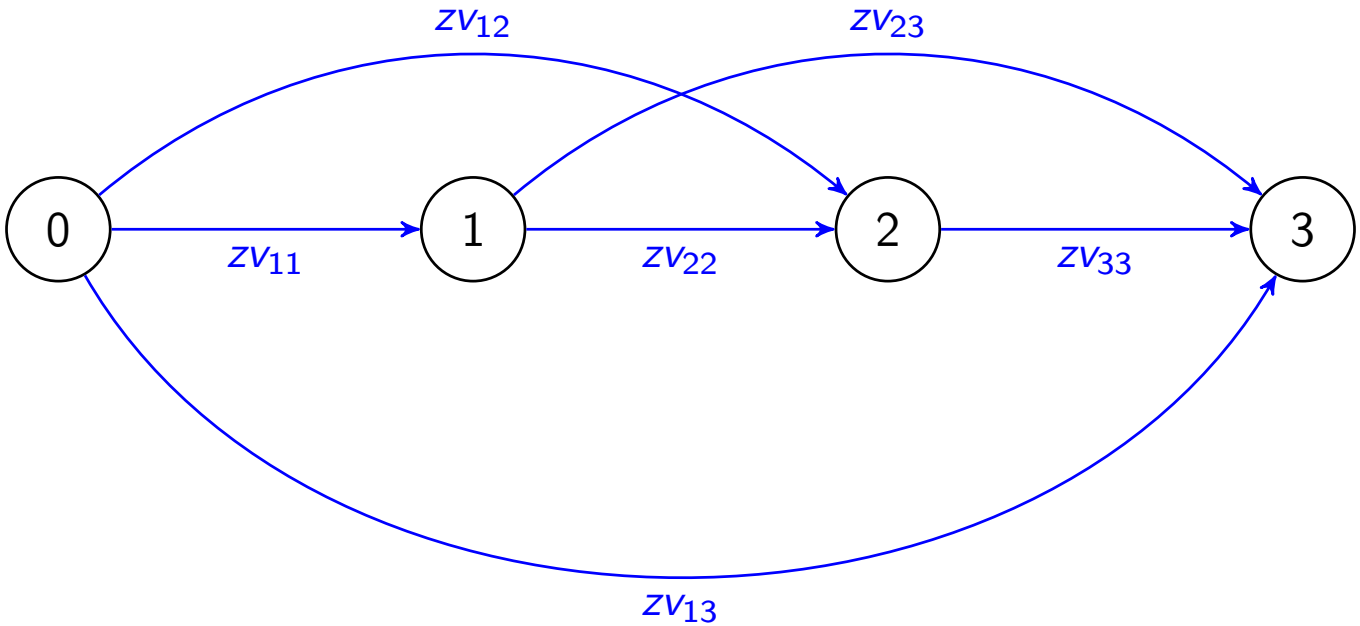
$$y_t \in \{0,1\}, \qquad \forall t \in T.$$

- The linear objective function minimizes the cost of the setups and of the arcs of the plan;
- constraints $\sum_k zv_{1k} = 1$ and $\sum_u zv_{u,t-1} - \sum_k zv_{tk} = 0$ are the flow constraints of a network (m linear constraints);
- constraints $\sum_k zv_{tk} \leq y_t$ force the setup if the product is produced in period t, and constraints $\sum_k D_{t,k}\, zv_{tk} \leq cap_t$ impose the capacities (m linear constraints each);
- finally, the domain constraints define the variables of the MIP model.

This MIP model is referred to as the $\mathrm{MIP}^{\mathrm{SP}}_{\mathrm{CLSP}}$ model, or **shortest-path formulation**.

The network of the variables zv_{tk}

- The nodes are $0, 1, \ldots, m$, and the variable zv_{tk} is the arc from node $t-1$ to node k, with cost cv_{tk}.
- A unit of flow leaves node 0 (constraint $\sum_k zv_{1k} = 1$), and the flow is conserved at the other nodes (constraints $\sum_u zv_{u,t-1} - \sum_k zv_{tk} = 0$).



Network of the variables zv_{tk} for $m = 3$: the variable zv_{tk} is the arc from node $t-1$ to node k.

Paths and plans

- A path from node 0 to node m is a plan that produces only when the stock is empty: each arc $(t-1, k)$ is a production in period t for the periods $t, \ldots, k$.
- This is why the formulation is called a **shortest-path formulation**: with the capacities relaxed, the optimal plan is a shortest path.

Setups, quantities and combinations of paths

- Constraints $\sum_k zv_{tk} \leq y_t$ link the arcs with the setups with the coefficient 1 instead of a big-M value.
- In constraints $\sum_k D_{t,k}\, zv_{tk} \leq cap_t$ the quantity produced in period t is $x_t = \sum_k D_{t,k}\, zv_{tk}$.
- The paths are the building blocks, and a plan that is not a path is a **combination of paths**: the quantities are integer, but the variables zv_{tk} may be fractional.

Example: $\mathrm{MIP}^{\mathrm{SP}}_{\mathrm{CLSP}}$ model and combinations of paths (1/6)

For the CLS reference instance the demands $D_{t,k}$ and the costs cv_{tk} are (the production cost is zero, so cv_{tk} is only holding cost):

	$D_{t,k}$					cv_{tk}				
$t \backslash k$	1	2	3	4	5	1	2	3	4	5
1	2	3	6	10	11	0	10	70	190	230
2		1	4	8	9		0	30	110	140
3			3	7	8			0	40	60
4				4	5				0	10
5					1					0

Example: $\mathrm{MIP}^{\mathrm{SP}}_{\mathrm{CLSP}}$ model and combinations of paths (2/6)

The $\mathrm{MIP}^{\mathrm{SP}}_{\mathrm{CLSP}}$ model for this instance is:

$$
\begin{aligned}
\text{min} \quad & 50\,y_1 + 50\,y_2 + 50\,y_3 + 50\,y_4 + 50\,y_5 \\
& +10\,zv_{12} + 70\,zv_{13} + 190\,zv_{14} + 230\,zv_{15} \\
& +30\,zv_{23} + 110\,zv_{24} + 140\,zv_{25} + 40\,zv_{34} \\
& +60\,zv_{35} + 10\,zv_{45}
\end{aligned}
$$

$$
\begin{aligned}
\text{subject to} \quad & zv_{11} + zv_{12} + zv_{13} + zv_{14} + zv_{15} = 1 \\
& zv_{11} - zv_{22} - zv_{23} - zv_{24} - zv_{25} = 0 \\
& zv_{12} + zv_{22} - zv_{33} - zv_{34} - zv_{35} = 0 \\
& zv_{13} + zv_{23} + zv_{33} - zv_{44} - zv_{45} = 0 \\
& zv_{14} + zv_{24} + zv_{34} + zv_{44} - zv_{55} = 0
\end{aligned}
$$

Example: $\mathrm{MIP}^{\mathrm{SP}}_{\mathrm{CLSP}}$ model and combinations of paths (3/6)

$$
\begin{aligned}
zv_{11} + zv_{12} + zv_{13} + zv_{14} + zv_{15} - y_1 &\leq 0 \\
zv_{22} + zv_{23} + zv_{24} + zv_{25} - y_2 &\leq 0 \\
zv_{33} + zv_{34} + zv_{35} - y_3 &\leq 0 \\
zv_{44} + zv_{45} - y_4 &\leq 0 \\
zv_{55} - y_5 &\leq 0 \\
2\,zv_{11} + 3\,zv_{12} + 6\,zv_{13} + 10\,zv_{14} + 11\,zv_{15} &\leq 4 \\
zv_{22} + 4\,zv_{23} + 8\,zv_{24} + 9\,zv_{25} &\leq 4 \\
3\,zv_{33} + 7\,zv_{34} + 8\,zv_{35} &\leq 4 \\
4\,zv_{44} + 5\,zv_{45} &\leq 4 \\
zv_{55} &\leq 4
\end{aligned}
$$

Example: $\mathrm{MIP}^{\mathrm{SP}}_{\mathrm{CLSP}}$ model and combinations of paths (4/6)

$$\begin{array}{lll} zv_{11},\ zv_{12},\ zv_{13},\ \ldots,\ zv_{55} & \geq & 0 \\ y_1,\ y_2,\ y_3,\ y_4,\ y_5 & \in & \{0,1\} \end{array}$$

An optimal solution $(\tilde{\boldsymbol{zv}}, \tilde{\boldsymbol{y}})(\mathrm{MIP}^{\mathrm{SP}}_{\mathrm{CLSP}})$ is:

$$\tilde{zv}_{12} = 1, \qquad \tilde{zv}_{33} = \tfrac{4}{5}, \quad \tilde{zv}_{35} = \tfrac{1}{5}, \qquad \tilde{zv}_{45} = \tfrac{4}{5},$$

$$\tilde{\boldsymbol{y}} = (1, 0, 1, 1, 0),$$

and all the other variables zv_{tk} are zero.

Example: $\mathrm{MIP}^{\mathrm{SP}}_{\mathrm{CLSP}}$ model and combinations of paths (5/6)

- It is the optimal plan $\tilde{\boldsymbol{x}} = (3, 0, 4, 4, 0)$, which is not a path: in period 3 it produces 4 units, more than the demand 3 of period 3, and less than the demand 7 of the periods 3 and 4.
- It is the combination, with weights $\frac{4}{5}$ and $\frac{1}{5}$, of the paths

$$0 \to 2 \to 3 \to 5 \quad (\text{plan } (3, 0, 3, 5, 0)) \quad \text{and}$$

$$0 \to 2 \to 5 \quad (\text{plan } (3, 0, 8, 0, 0)).$$

- The quantities are $x_1 = D_{1,2}\, zv_{12} = 3$, $x_3 = 3 \cdot \frac{4}{5} + 8 \cdot \frac{1}{5} = 4$ and $x_4 = 5 \cdot \frac{4}{5} = 4$.

Example: $\mathrm{MIP}^{\mathrm{SP}}_{\mathrm{CLSP}}$ model and combinations of paths (6/6)

The optimal value $z(\mathrm{MIP}^{\mathrm{SP}}_{\mathrm{CLSP}})$ is

$$\underbrace{3 \cdot 50}_{\text{setups}} + \underbrace{10 \cdot 1 + 0 \cdot \tfrac{4}{5} + 60 \cdot \tfrac{1}{5} + 10 \cdot \tfrac{4}{5}}_{=30} = 180.$$

Relaxing the setup variables

- To obtain an LP relaxation of the $\mathrm{MIP}^{\mathrm{SP}}_{\mathrm{CLSP}}$ model, we relax constraints $y_t \in \{0, 1\}$ into the bounds $0 \leq y_t \leq 1$.
- The obtained LP model is written in the next slides.

The LP relaxation of the shortest-path formulation

$$\min \quad \sum_{t \in T} sc_t \, y_t + \sum_{t \in T} \sum_{k=t}^{m} cv_{tk} \, zv_{tk}$$

$$\text{subject to} \quad \sum_{k=1}^{m} zv_{1k} = 1,$$

$$\sum_{u=1}^{t-1} zv_{u,t-1} - \sum_{k=t}^{m} zv_{tk} = 0, \qquad \forall t \in T \setminus \{1\},$$

$$\sum_{k=t}^{m} zv_{tk} \leq y_t, \qquad \forall t \in T,$$

$$\sum_{k=t}^{m} D_{t,k} \, zv_{tk} \leq cap_t, \qquad \forall t \in T,$$

The model $\mathrm{LP}^{\mathrm{SP}}_{\mathrm{CLSP}}$: bounds and domains

$$
\begin{aligned}
y_t &\leq 1, && \forall t \in T, \\
zv_{tk} &\geq 0, && \forall t, k \in T,\ k \geq t, \\
y_t &\geq 0, && \forall t \in T.
\end{aligned}
$$

- This LP model is referred to as the $\mathrm{LP}^{\mathrm{SP}}_{\mathrm{CLSP}}$ model.
- It differs from the $\mathrm{MIP}^{\mathrm{SP}}_{\mathrm{CLSP}}$ model only in the bounds $0 \leq y_t \leq 1$ in place of the binary constraints $y_t \in \{0, 1\}$.

Example: the $\mathrm{LP}^{\mathrm{SP}}_{\mathrm{CLSP}}$ model (1/4)

The $\mathrm{LP}^{\mathrm{SP}}_{\mathrm{CLSP}}$ model for the CLS reference instance is:

$$
\begin{aligned}
\min \quad & 50\,y_1 + 50\,y_2 + 50\,y_3 + 50\,y_4 + 50\,y_5 \\
& +10\,zv_{12} + 70\,zv_{13} + 190\,zv_{14} + 230\,zv_{15} \\
& +30\,zv_{23} + 110\,zv_{24} + 140\,zv_{25} + 40\,zv_{34} \\
& +60\,zv_{35} + 10\,zv_{45}
\end{aligned}
$$

$$
\begin{aligned}
\text{subject to} \quad & zv_{11} + zv_{12} + zv_{13} + zv_{14} + zv_{15} = 1 \\
& zv_{11} - zv_{22} - zv_{23} - zv_{24} - zv_{25} = 0 \\
& zv_{12} + zv_{22} - zv_{33} - zv_{34} - zv_{35} = 0 \\
& zv_{13} + zv_{23} + zv_{33} - zv_{44} - zv_{45} = 0 \\
& zv_{14} + zv_{24} + zv_{34} + zv_{44} - zv_{55} = 0
\end{aligned}
$$

Example: the $\mathrm{LP}^{\mathrm{SP}}_{\mathrm{CLSP}}$ model (2/4)

$$
\begin{aligned}
zv_{11} + zv_{12} + zv_{13} + zv_{14} + zv_{15} - y_1 &\leq 0 \\
zv_{22} + zv_{23} + zv_{24} + zv_{25} - y_2 &\leq 0 \\
zv_{33} + zv_{34} + zv_{35} - y_3 &\leq 0 \\
zv_{44} + zv_{45} - y_4 &\leq 0 \\
zv_{55} - y_5 &\leq 0 \\
2\,zv_{11} + 3\,zv_{12} + 6\,zv_{13} + 10\,zv_{14} + 11\,zv_{15} &\leq 4 \\
zv_{22} + 4\,zv_{23} + 8\,zv_{24} + 9\,zv_{25} &\leq 4 \\
3\,zv_{33} + 7\,zv_{34} + 8\,zv_{35} &\leq 4 \\
4\,zv_{44} + 5\,zv_{45} &\leq 4 \\
zv_{55} &\leq 4
\end{aligned}
$$

Example: the $\mathrm{LP}^{\mathrm{SP}}_{\mathrm{CLSP}}$ model (3/4)

$$
\begin{aligned}
y_1 &\leq 1 \\
y_2 &\leq 1 \\
y_3 &\leq 1 \\
y_4 &\leq 1 \\
y_5 &\leq 1 \\
zv_{11},\ zv_{12},\ zv_{13},\ \ldots,\ zv_{55} &\geq 0 \\
y_1,\ y_2,\ y_3,\ y_4,\ y_5 &\geq 0
\end{aligned}
$$

Example: the $\mathrm{LP}^{\mathrm{SP}}_{\mathrm{CLSP}}$ model (4/4)

An optimal solution $(\tilde{\boldsymbol{zv}}, \tilde{\boldsymbol{y}})(\mathrm{LP}^{\mathrm{SP}}_{\mathrm{CLSP}})$ is:

$$\tilde{zv}_{12} = 1, \qquad \tilde{zv}_{33} = \tfrac{4}{5}, \quad \tilde{zv}_{35} = \tfrac{1}{5}, \qquad \tilde{zv}_{45} = \tfrac{4}{5},$$

$$\tilde{\boldsymbol{y}} = (1, 0, 1, \tfrac{4}{5}, 0),$$

and all the other variables zv_{tk} are zero.

- These are the arcs of the optimal plan, with the setup of period 4 paid only for the fraction $\frac{4}{5}$.
- The optimal value $z(\mathrm{LP}^{\mathrm{SP}}_{\mathrm{CLSP}})$ is $180 - \frac{1}{5} \cdot 50 = 170$.

Summary

The shortest-path formulation $\mathrm{MIP}^{\mathrm{SP}}_{\mathrm{CLSP}}$:

- Continuous variables zv_{tk}, the arcs of a network with nodes $0, 1, \ldots, m$, with costs cv_{tk}
- A path from 0 to m is a plan; a plan that is not a path is a combination of paths
- The link $\sum_k zv_{tk} \leq y_t$ has coefficient 1, with no big-M value

The reference instance:

- The optimal plan $(3, 0, 4, 4, 0)$, of cost 180, combines two paths
- The LP relaxation $\mathrm{LP}^{\mathrm{SP}}_{\mathrm{CLSP}}$ gives the bound 170

Comparing the formulations

Introduction: comparing the formulations

- We put side by side the lower bounds of all the relaxations computed so far
- A stronger bound is closer to the optimal value, but it may come from a larger model
- The comparison shows what we gain and what we pay with each formulation

Same optimal value, different bounds

- The formulations have the same optimal value $z(\mathrm{ILP}_{\mathrm{CLSP}}) = z(\mathrm{MIP}^{\mathrm{SP}}_{\mathrm{CLSP}}) = 180$, but their LP relaxations have different optimal values.
- On the reference instance the lower bounds, computed with the exact simplex algorithm and checked with Gurobi, are given in the next slide.
- The last column is the gap of the chapter «Combinatorial optimization problems» between the bound and the optimal value; it is undefined for the bound 0.

Lower bounds on the reference instance

Model	Lower bound	Gap $\frac{180-\mathrm{LB}}{\mathrm{LB}}$
Without setups	0	undefined
$\mathrm{LP}^{\mathrm{dem}}_{\mathrm{CLSP}}$ ($M_t = D_{t,m}$)	99.65	80.6%
$\mathrm{LP}^{\mathrm{cap}}_{\mathrm{CLSP}}$ ($M_t = cap_t$)	137.50	30.9%
$\mathrm{LP}^{\mathrm{strong}}_{\mathrm{CLSP}}$ ($M_t = \min\{cap_t, D_{t,m}\}$)	157.50	14.3%
$\mathrm{LP}^{\mathrm{SP}}_{\mathrm{CLSP}}$ (shortest path)	170	5.9%

Strength and size

- The bound 170 of the shortest-path formulation is the optimal value of the problem without capacity (plan $(3, 0, 8, 0, 0)$, of cost 170): the capacity constraints $\sum_k D_{t,k}\, zv_{tk} \leq cap_t$ act only on the quantities, and a combination of paths can satisfy them.

- The price of the shortest-path formulation is its size: there are $O(m^2)$ variables zv_{tk}, against the $O(m)$ variables of the big-M formulation.

Summary

- The two formulations have the same optimal value 180, but different LP bounds
- On the reference instance the bound grows from 99.65 (big-M with $D_{t,m}$) to 157.5 (strong big-M) and to 170 (shortest path)
- The shortest-path formulation is stronger but larger: $O(m^2)$ variables against $O(m)$

ILP models for the variants

Introduction: ILP models for the variants

- The problems of this section are the variants of the capacitated lot sizing problem defined in the chapter «Lot sizing problems»
- We write the ILP model of each variant, in the order of that chapter:
  - the uncapacitated lot sizing problem;
  - several products and setup times;
  - initial stock;
  - overtime;
  - unsatisfied demand.
- Each model keeps the structure of the big-M model of the CLS

The uncapacitated lot sizing problem: variables

- Given a set $T = \{1, 2, \ldots, m\}$ of periods with demands d_t and no binding capacity, we introduce, for every period $t \in T$, the following **integer variables** and **binary variables**:

$$x_t \in \mathbb{Z}_{\geq 0} \text{ (production)},$$
$$s_t \in \mathbb{Z}_{\geq 0} \text{ (inventory at the end of the period)},$$

$$y_t = \begin{cases} 1 & \text{if there is a setup in period } t \\ 0 & \text{otherwise} \end{cases}$$

- Since $s_m = 0$, the production of period t is at most the demand $D_{t,m}$ of the periods $t, \ldots, m$: this is the big-M value of the link between x_t and y_t.

The model $\mathrm{ILP}_{\mathrm{ULS}}$

Using these variables, an Integer Linear Programming (ILP) model for the uncapacitated lot sizing problem (ULS) reads as follows:

$$\min \quad \sum_{t \in T} (sc_t\, y_t + vc_t\, x_t + hc_t\, s_t)$$

$$\begin{aligned}
\text{subject to} \quad & s_{t-1} + x_t = d_t + s_t, && \forall t \in T, \\
& x_t \leq D_{t,m}\, y_t, && \forall t \in T, \\
& s_0 = s_m = 0, && \\
& x_t \in \mathbb{Z}_{\geq 0}, && \forall t \in T, \\
& s_t \in \mathbb{Z}_{\geq 0}, && \forall t \in T, \\
& y_t \in \{0, 1\}, && \forall t \in T.
\end{aligned}$$

This ILP model is referred to as the $\mathrm{ILP}_{\mathrm{ULS}}$ model.

Description of the model $\mathrm{ILP}_{\mathrm{ULS}}$

- The linear objective function minimizes the cost of the setups, of the production and of the holding;
- constraints $s_{t-1} + x_t = d_t + s_t$ are the flow balancing constraints (m linear constraints);
- constraints $x_t \leq D_{t,m}\, y_t$ force a setup in every period with positive production (m linear constraints);
- constraint $s_0 = s_m = 0$ imposes an empty inventory at the beginning and at the end of the horizon;
- finally, the domain constraints define the variables of the ILP model.

The ULS and the CLS

- The $\mathrm{ILP}_{\mathrm{ULS}}$ model is the $\mathrm{ILP}_{\mathrm{CLSP}}$ model without the capacity constraints $x_t \leq cap_t$ and with the big-M values $M_t = D_{t,m}$.
- The DP algorithm of Wagner and Whitin, in the chapter «DP for the uncapacitated lot sizing problem» later in the course, solves the ULS in polynomial time.

Example: the $\mathrm{ILP}_{\mathrm{ULS}}$ model (1/3)

We consider the CLS reference instance without the capacities: the demands are $\boldsymbol{d} = (2, 1, 3, 4, 1)$, the setup costs $sc_t = 50$, the holding costs $hc_t = 10$, and $vc_t = 0$. The big-M values are $D_{t,5} = (11, 9, 8, 5, 1)$, and the $\mathrm{ILP}_{\mathrm{ULS}}$ model for this ULS instance is:

$$
\begin{aligned}
\min \quad & 50\,y_1 + 50\,y_2 + 50\,y_3 + 50\,y_4 + 50\,y_5 + 10\,s_1 \\
& +10\,s_2 + 10\,s_3 + 10\,s_4 + 10\,s_5 \\
\text{subject to} \quad & x_1 - s_1 = 2 \\
& s_1 + x_2 - s_2 = 1 \\
& s_2 + x_3 - s_3 = 3 \\
& s_3 + x_4 - s_4 = 4 \\
& s_4 + x_5 - s_5 = 1
\end{aligned}
$$

Example: the $\mathrm{ILP}_{\mathrm{ULS}}$ model (2/3)

$$
\begin{aligned}
x_1 - 11\, y_1 &\leq 0 \\
x_2 - 9\, y_2 &\leq 0 \\
x_3 - 8\, y_3 &\leq 0 \\
x_4 - 5\, y_4 &\leq 0 \\
x_5 - y_5 &\leq 0 \\
s_5 &= 0 \\
\\
x_1,\ x_2,\ x_3,\ x_4,\ x_5 &\in \mathbb{Z}_{\geq 0} \\
s_1,\ s_2,\ s_3,\ s_4,\ s_5 &\in \mathbb{Z}_{\geq 0} \\
y_1,\ y_2,\ y_3,\ y_4,\ y_5 &\in \{0, 1\}
\end{aligned}
$$

Example: the $\mathrm{ILP}_{\mathrm{ULS}}$ model (3/3)

An optimal solution $(\tilde{\boldsymbol{y}}, \tilde{\boldsymbol{x}}, \tilde{\boldsymbol{s}})(\mathrm{ILP}_{\mathrm{ULS}})$ is:

$$\tilde{\boldsymbol{y}} = (1, 0, 1, 0, 0), \qquad \tilde{\boldsymbol{x}} = (3, 0, 8, 0, 0), \qquad \tilde{\boldsymbol{s}} = (1, 0, 5, 1, 0)$$

- The optimal value $z(\mathrm{ILP}_{\mathrm{ULS}})$ is $2 \cdot 50 + (1 + 5 + 1) \cdot 10 = 170$, the bound of the shortest-path formulation of the CLS.
- The optimal solution is not unique: the plan $(3, 0, 3, 5, 0)$, with three setups and the stocks $(1, 0, 0, 1, 0)$, also costs $3 \cdot 50 + 2 \cdot 10 = 170$.

Several products and setup times: data

- A set P of products shares the resource of each period.
- We write d_{pt}, sc_{pt}, vc_{pt} and hc_{pt} for the data of product $p \in P$ in period $t \in T$.
- Producing a unit of p takes the time $vt_{pt} \in \mathbb{Z}_{\geq 1}$, a setup takes the time $st_{pt} \in \mathbb{Z}_{\geq 0}$, and the capacity cap_t is a time.
- For $j \leq k$ we write $sd_{pjk} = \sum_{u=j}^{k} d_{pu}$ for the demand of product p in the periods $j, \ldots, k$.

Several products and setup times: variables

- We introduce, for every product $p \in P$ and every period $t \in T$, the following **integer variables** and **binary variables**:

$$x_{pt} \in \mathbb{Z}_{\geq 0} \text{ (production)},$$
$$s_{pt} \in \mathbb{Z}_{\geq 0} \text{ (inventory at the end of the period)},$$

$$y_{pt} = \begin{cases} 1 & \text{if there is a setup of product } p \text{ in period } t \\ 0 & \text{otherwise} \end{cases}$$

- The inventories at the beginning of the horizon are the constants $s_{p0} = 0$.

The model $\mathrm{ILP}^{P}_{\mathrm{CLSP}}$: objective and constraints

Using these variables, an Integer Linear Programming (ILP) model for the lot sizing problem with several products and setup times (MLS) reads as follows:

$$\min \quad \sum_{p \in P} \sum_{t \in T} (sc_{pt}\, y_{pt} + vc_{pt}\, x_{pt} + hc_{pt}\, s_{pt})$$

$$\text{subject to} \quad s_{p,t-1} + x_{pt} = d_{pt} + s_{pt}, \quad \forall p \in P,\ \forall t \in T,$$

$$\sum_{p \in P} (st_{pt}\, y_{pt} + vt_{pt}\, x_{pt}) \leq cap_t, \qquad \forall t \in T,$$

$$x_{pt} \leq M_{pt}\, y_{pt}, \qquad \forall p \in P,\ \forall t \in T,$$

$$s_{p0} = s_{pm} = 0, \qquad \forall p \in P,$$

The model $\mathrm{ILP}^{P}_{\mathrm{CLSP}}$: domains of the variables

$$x_{pt} \in \mathbb{Z}_{\geq 0}, \quad \forall p \in P,\ \forall t \in T,$$

$$s_{pt} \in \mathbb{Z}_{\geq 0}, \quad \forall p \in P,\ \forall t \in T,$$

$$y_{pt} \in \{0,1\}, \quad \forall p \in P,\ \forall t \in T.$$

This ILP model is referred to as the $\mathrm{ILP}^{P}_{\mathrm{CLSP}}$ model.

Description of the model $\mathrm{ILP}^{P}_{\mathrm{CLSP}}$

- The linear objective function minimizes the cost of the setups, of the production and of the holding of all the products;
- constraints $s_{p,t-1} + x_{pt} = d_{pt} + s_{pt}$ are the flow balancing constraints of every product ($|P|\, m$ linear constraints);
- constraints $\sum_p (st_{pt}\, y_{pt} + vt_{pt}\, x_{pt}) \leq cap_t$ impose that the setup times and the production times of each period do not exceed its capacity (m linear constraints);
- constraints $x_{pt} \leq M_{pt}\, y_{pt}$ link the quantities with the setups ($|P|\, m$ linear constraints);
- constraints $s_{p0} = s_{pm} = 0$ impose an empty inventory at the beginning and at the end of the horizon;
- finally, the domain constraints define the variables of the ILP model.

Strong big-M values and products

- A setup of p in period t leaves the time $cap_t - st_{pt}$ for the production, and $s_{pm} = 0$ bounds the production by the remaining demand. The strong big-M value is

$$M_{pt} = \min \left\{ \left\lfloor \frac{cap_t - st_{pt}}{vt_{pt}} \right\rfloor, \; sd_{ptm} \right\}.$$

- The products interact only through the capacity constraints $\sum_p (st_{pt}\, y_{pt} + vt_{pt}\, x_{pt}) \leq cap_t$: without them, the model splits into $|P|$ independent single-product models.

Example: the $\mathrm{ILP}^{P}_{\mathrm{CLSP}}$ model (1/5)

We consider $|P| = 2$ products and $m = 3$ periods, with capacities $cap_t = 9$, production times $vt_{pt} = 1$, setup times $st_{pt} = 1$, holding costs $hc_{pt} = 5$ and production costs $vc_{pt} = 0$. The setup costs are $sc_{1t} = 30$ and $sc_{2t} = 40$, and the demands are:

	product 1			product 2		
period t	1	2	3	1	2	3
d_{pt}	3	2	4	2	4	1
sd_{ptm}	9	6	4	7	5	1
M_{pt}	8	6	4	7	5	1

Example: the $\mathrm{ILP}^{P}_{\mathrm{CLSP}}$ model (2/5)

The $\mathrm{ILP}^{P}_{\mathrm{CLSP}}$ model with the strong big-M values for this instance is:

$$
\begin{aligned}
\text{min} \quad & 30\,y_{11} + 30\,y_{12} + 30\,y_{13} + 40\,y_{21} + 40\,y_{22} + 40\,y_{23} \\
& +5\,s_{11} + 5\,s_{12} + 5\,s_{13} + 5\,s_{21} + 5\,s_{22} + 5\,s_{23}
\end{aligned}
$$

$$
\begin{aligned}
\text{subject to} \quad & x_{11} - s_{11} && = 3 \\
& s_{11} + x_{12} - s_{12} && = 2 \\
& s_{12} + x_{13} - s_{13} && = 4 \\
& x_{21} - s_{21} && = 2 \\
& s_{21} + x_{22} - s_{22} && = 4 \\
& s_{22} + x_{23} - s_{23} && = 1
\end{aligned}
$$

Example: the $\mathrm{ILP}^{P}_{\mathrm{CLSP}}$ model (3/5)

$$
\begin{aligned}
x_{11} + x_{21} + y_{11} + y_{21} &\leq 9 \\
x_{12} + x_{22} + y_{12} + y_{22} &\leq 9 \\
x_{13} + x_{23} + y_{13} + y_{23} &\leq 9 \\
x_{11} - 8\, y_{11} &\leq 0 \\
x_{12} - 6\, y_{12} &\leq 0 \\
x_{13} - 4\, y_{13} &\leq 0 \\
x_{21} - 7\, y_{21} &\leq 0 \\
x_{22} - 5\, y_{22} &\leq 0 \\
x_{23} - y_{23} &\leq 0
\end{aligned}
$$

Example: the $\mathrm{ILP}^{P}_{\mathrm{CLSP}}$ model (4/5)

$$
\begin{aligned}
s_{13} &= 0 \\
s_{23} &= 0 \\
x_{11},\ x_{12},\ x_{13},\ x_{21},\ x_{22},\ x_{23} &\in \mathbb{Z}_{\geq 0} \\
s_{11},\ s_{12},\ s_{13},\ s_{21},\ s_{22},\ s_{23} &\in \mathbb{Z}_{\geq 0} \\
y_{11},\ y_{12},\ y_{13},\ y_{21},\ y_{22},\ y_{23} &\in \{0,1\}
\end{aligned}
$$

An optimal solution $(\tilde{\boldsymbol{y}}, \tilde{\boldsymbol{x}}, \tilde{\boldsymbol{s}})(\mathrm{ILP}^{P}_{\mathrm{CLSP}})$ is:

$$
\begin{aligned}
&\tilde{y}_{11} = \tilde{y}_{13} = \tilde{y}_{21} = \tilde{y}_{22} = 1, \quad \tilde{y}_{12} = \tilde{y}_{23} = 0, \\
&\tilde{x}_{11} = 5, \quad \tilde{x}_{13} = 4, \quad \tilde{x}_{21} = 2, \quad \tilde{x}_{22} = 5, \quad \tilde{x}_{12} = \tilde{x}_{23} = 0, \\
&\tilde{s}_{11} = 2, \quad \tilde{s}_{22} = 1, \quad \text{all the other } \tilde{s}_{pt} = 0
\end{aligned}
$$

Example: the $\mathrm{ILP}^{P}_{\mathrm{CLSP}}$ model (5/5)

- The optimal value $z(\mathrm{ILP}^{P}_{\mathrm{CLSP}})$ is $2 \cdot 30 + 2 \cdot 40 + (2 + 1) \cdot 5 = 155$, and the optimal solution is unique (Gurobi).
- Period 1 uses its whole capacity, $5 + 2 + 1 + 1 = 9$.
- Without the capacity constraints product 2 would be produced only in period 1, with value 140: this plan needs the time $5 + 7 + 1 + 1 = 14 > 9$.

Initial stock: data and variables

- The data are those of the lot sizing problem with several products and setup times. In addition, the initial inventory of product p is a decision, bought at the cost fc_p per unit.

- We introduce the integer variables x_{pt}, s_{pt} and the binary variables y_{pt} of the $\mathrm{ILP}^P_{\mathrm{CLSP}}$ model, and, for every product $p \in P$, the following **integer variable**:

$$si_p \in \mathbb{Z}_{\geq 0} \text{ (inventory of product } p \text{ at the beginning of the horizon).}$$

The model $\mathrm{ILP}^{\mathrm{SI}}_{\mathrm{CLSP}}$: objective and constraints

Using these variables, an Integer Linear Programming (ILP) model for the lot sizing problem with initial stock (LS-SI) reads as follows:

$$\min \quad \sum_{p \in P} fc_p \, si_p + \sum_{p \in P} \sum_{t \in T} (sc_{pt} \, y_{pt} + vc_{pt} \, x_{pt} + hc_{pt} \, s_{pt})$$

$$\text{subject to} \quad si_p + x_{p1} = d_{p1} + s_{p1}, \qquad \forall p \in P,$$

$$s_{p,t-1} + x_{pt} = d_{pt} + s_{pt}, \quad \forall p \in P, \ \forall t \in T \setminus \{1\},$$

$$\sum_{p \in P} (st_{pt} \, y_{pt} + vt_{pt} \, x_{pt}) \leq cap_t, \qquad \forall t \in T,$$

$$x_{pt} \leq M_{pt} \, y_{pt}, \qquad \forall p \in P, \ \forall t \in T,$$

$$s_{pm} = 0, \qquad \forall p \in P,$$

The model $\mathrm{ILP}^{\mathrm{SI}}_{\mathrm{CLSP}}$: domains of the variables

$$
\begin{aligned}
x_{pt} &\in \mathbb{Z}_{\geq 0}, && \forall p \in P,\ \forall t \in T,\\
s_{pt} &\in \mathbb{Z}_{\geq 0}, && \forall p \in P,\ \forall t \in T,\\
si_{p} &\in \mathbb{Z}_{\geq 0}, && \forall p \in P,\\
y_{pt} &\in \{0,1\}, && \forall p \in P,\ \forall t \in T.
\end{aligned}
$$

- This ILP model is referred to as the $\mathrm{ILP}^{\mathrm{SI}}_{\mathrm{CLSP}}$ model.
- It is the $\mathrm{ILP}^{P}_{\mathrm{CLSP}}$ model with the variables si_p in place of the constants $s_{p0} = 0$, and with their cost in the objective function; the strong big-M values M_{pt} are the same.

Description of the model $\mathrm{ILP}^{\mathrm{SI}}_{\mathrm{CLSP}}$

- The linear objective function minimizes the cost of the initial inventory, of the setups, of the production and of the holding;
- constraints $si_p + x_{p1} = d_{p1} + s_{p1}$ and $s_{p,t-1} + x_{pt} = d_{pt} + s_{pt}$ are the flow balancing constraints of every product, with the initial inventory si_p in the first period ($|P|\, m$ linear constraints);
- constraints $\sum_p (st_{pt}\, y_{pt} + vt_{pt}\, x_{pt}) \leq cap_t$ impose that the setup times and the production times of each period do not exceed its capacity (m linear constraints); constraints $x_{pt} \leq M_{pt}\, y_{pt}$ link the quantities with the setups ($|P|\, m$ linear constraints);
- constraints $s_{pm} = 0$ impose an empty inventory at the end of the horizon ($|P|$ linear constraints);
- finally, the domain constraints define the variables of the ILP model.

Example: the $\mathrm{ILP}^{\mathrm{SI}}_{\mathrm{CLSP}}$ model (1/6)

We consider $|P| = 2$ products and $m = 3$ periods, with capacities $cap_1 = 2500$, $cap_2 = 2100$, $cap_3 = 2500$, the initial inventory costs $fc_1 = fc_2 = 3000$ and the setup costs $sc_{pt} = 10000$. The other data are:

	product 1			product 2		
period t	1	2	3	1	2	3
d_{pt}	100	200	10	10	100	150
vt_{pt}	20	10	10	10	5	5
st_{pt}	100	100	100	110	110	110
hc_{pt}	110	100	100	105	105	105
vc_{pt}	1100	1000	1000	900	800	800

Example: the $\mathrm{ILP}^{\mathrm{SI}}_{\mathrm{CLSP}}$ model (2/6)

The big-M values are:

period t	1	2	3	1	2	3
sd_{ptm}	310	210	10	260	250	150
$\lfloor (cap_t - st_{pt})/vt_{pt} \rfloor$	120	200	240	239	398	478
M_{pt}	120	200	10	239	250	150

Example: the $\mathrm{ILP}^{\mathrm{SI}}_{\mathrm{CLSP}}$ model (3/6)

The $\mathrm{ILP}^{\mathrm{SI}}_{\mathrm{CLSP}}$ model with the strong big-M values for this instance is:

$$
\begin{aligned}
\min \quad & 3000\, si_1 + 3000\, si_2 + 10000\, y_{11} + 10000\, y_{12} \\
& +10000\, y_{13} + 10000\, y_{21} + 10000\, y_{22} + 10000\, y_{23} \\
& +1100\, x_{11} + 1000\, x_{12} + 1000\, x_{13} + 900\, x_{21} \\
& +800\, x_{22} + 800\, x_{23} + 110\, s_{11} + 100\, s_{12} \\
& +100\, s_{13} + 105\, s_{21} + 105\, s_{22} + 105\, s_{23}
\end{aligned}
$$

$$
\begin{aligned}
\text{subject to} \quad & si_1 + x_{11} - s_{11} && = && 100 \\
& s_{11} + x_{12} - s_{12} && = && 200 \\
& s_{12} + x_{13} - s_{13} && = && 10 \\
& si_2 + x_{21} - s_{21} && = && 10 \\
& s_{21} + x_{22} - s_{22} && = && 100 \\
& s_{22} + x_{23} - s_{23} && = && 150
\end{aligned}
$$

Example: the $\mathrm{ILP}^{\mathrm{SI}}_{\mathrm{CLSP}}$ model (4/6)

$$
\begin{aligned}
20\,x_{11} + 10\,x_{21} + 100\,y_{11} + 110\,y_{21} &\leq 2500 \\
10\,x_{12} + 5\,x_{22} + 100\,y_{12} + 110\,y_{22} &\leq 2100 \\
10\,x_{13} + 5\,x_{23} + 100\,y_{13} + 110\,y_{23} &\leq 2500 \\
x_{11} - 120\,y_{11} &\leq 0 \\
x_{12} - 200\,y_{12} &\leq 0 \\
x_{13} - 10\,y_{13} &\leq 0 \\
x_{21} - 239\,y_{21} &\leq 0 \\
x_{22} - 250\,y_{22} &\leq 0 \\
x_{23} - 150\,y_{23} &\leq 0
\end{aligned}
$$

Example: the $\mathrm{ILP}^{\mathrm{SI}}_{\mathrm{CLSP}}$ model (5/6)

$$
\begin{aligned}
s_{13} &= 0 \\
s_{23} &= 0 \\
si_1,\ si_2 &\in \mathbb{Z}_{\geq 0} \\
x_{11},\ x_{12},\ x_{13},\ x_{21},\ x_{22},\ x_{23} &\in \mathbb{Z}_{\geq 0} \\
s_{11},\ s_{12},\ s_{13},\ s_{21},\ s_{22},\ s_{23} &\in \mathbb{Z}_{\geq 0} \\
y_{11},\ y_{12},\ y_{13},\ y_{21},\ y_{22},\ y_{23} &\in \{0,1\}
\end{aligned}
$$

Example: the $\mathrm{ILP}^{\mathrm{SI}}_{\mathrm{CLSP}}$ model (6/6)

An optimal solution of the $\mathrm{ILP}^{\mathrm{SI}}_{\mathrm{CLSP}}$ model is:

$$\tilde{x}_{11} = 59, \quad \tilde{x}_{12} = 200, \quad \tilde{x}_{13} = 10,$$
$$\tilde{x}_{21} = 110, \quad \tilde{x}_{22} = 0, \quad \tilde{x}_{23} = 150,$$
$$\tilde{si}_1 = 41, \quad \tilde{si}_2 = 0, \quad \tilde{s}_{21} = 100, \quad \text{all the other } \tilde{s}_{pt} = 0,$$

with a setup in every period except for product 2 in period 2.

- The optimal value $z(\mathrm{ILP}^{\mathrm{SI}}_{\mathrm{CLSP}})$ is 677 400, and the optimal solution is unique (Gurobi).
- Product 1 starts with 41 units bought at the beginning of the horizon, because period 1 has no time to produce all its demand.

The LP relaxation $\mathrm{LP}^{\mathrm{SI}}_{\mathrm{CLSP}}$: objective and constraints

To obtain an LP relaxation of the $\mathrm{ILP}^{\mathrm{SI}}_{\mathrm{CLSP}}$ model, we relax the integrality constraints of the variables into non-negativity constraints and the bounds $y_{pt} \leq 1$. The obtained LP model reads as follows:

$$\min \quad \sum_{p \in P} fc_p \, si_p + \sum_{p \in P} \sum_{t \in T} (sc_{pt} \, y_{pt} + vc_{pt} \, x_{pt} + hc_{pt} \, s_{pt})$$

$$\text{subject to} \quad si_p + x_{p1} = d_{p1} + s_{p1}, \qquad \forall p \in P,$$

$$s_{p,t-1} + x_{pt} = d_{pt} + s_{pt}, \quad \forall p \in P, \ \forall t \in T \setminus \{1\},$$

$$\sum_{p \in P} (st_{pt} \, y_{pt} + vt_{pt} \, x_{pt}) \leq cap_t, \qquad \forall t \in T,$$

$$x_{pt} \leq M_{pt} \, y_{pt}, \qquad \forall p \in P, \ \forall t \in T,$$

$$s_{pm} = 0, \qquad \forall p \in P,$$

The LP relaxation $\mathrm{LP}^{\mathrm{SI}}_{\mathrm{CLSP}}$: bounds and domains

$$
\begin{aligned}
y_{pt} &\leq 1, && \forall p \in P,\ \forall t \in T, \\
x_{pt} &\geq 0, && \forall p \in P,\ \forall t \in T, \\
s_{pt} &\geq 0, && \forall p \in P,\ \forall t \in T, \\
y_{pt} &\geq 0, && \forall p \in P,\ \forall t \in T, \\
si_{p} &\geq 0, && \forall p \in P.
\end{aligned}
$$

This LP model is referred to as the $\mathrm{LP}^{\mathrm{SI}}_{\mathrm{CLSP}}$ model.

A shortest-path formulation with initial stock

- In the shortest-path formulation each product has its own network, with an additional continuous variable $w_{pt} \geq 0$, the arc from node 0 to node t, for the fraction of the initial inventory that covers the demands of the first t periods.

- The costs of the arcs are

$$cv_{ptk} = vc_{pt}\, sd_{ptk} + \sum_{s=t+1}^{k} \sum_{u=t}^{s-1} hc_{pu}\, d_{ps},$$

$$ci_{pt} = fc_{p}\, sd_{p1t} + \sum_{s=2}^{t} \sum_{u=1}^{s-1} hc_{pu}\, d_{ps}.$$

The model $\mathrm{MIP}^{\mathrm{SP,SI}}_{\mathrm{CLSP}}$: objective and constraints

Using these variables, a MIP model for the lot sizing problem with initial stock (LS-SI) reads as follows:

$$\min \quad \sum_{p\in P}\sum_{t\in T}(sc_{pt}\, y_{pt} + ci_{pt}\, w_{pt}) + \sum_{p\in P}\sum_{t\in T}\sum_{k=t}^{m} cv_{ptk}\, zv_{ptk}$$

$$\text{subject to} \quad \sum_{k=1}^{m}(w_{pk} + zv_{p1k}) = 1, \qquad \forall p \in P,$$

$$w_{p,t-1} + \sum_{u=1}^{t-1} zv_{p,u,t-1} - \sum_{k=t}^{m} zv_{ptk} = 0, \qquad \forall p \in P,\ \forall t \in T \setminus \{1\},$$

$$\sum_{k=t}^{m} zv_{ptk} \le y_{pt}, \qquad \forall p \in P,\ \forall t \in T,$$

$$\sum_{p\in P} st_{pt}\, y_{pt} + \sum_{p\in P}\sum_{k=t}^{m} vt_{pt}\, sd_{ptk}\, zv_{ptk} \le cap_t, \qquad \forall t \in T,$$

The model $\mathrm{MIP}^{\mathrm{SP,SI}}_{\mathrm{CLSP}}$: integrality and domains

$$\sum_{k=t}^{m} sd_{ptk}\, zv_{ptk} \in \mathbb{Z}_{\geq 0}, \qquad \forall p \in P,\ \forall t \in T,$$

$$\sum_{t \in T} sd_{p1t}\, w_{pt} \in \mathbb{Z}_{\geq 0}, \qquad \forall p \in P,$$

$$zv_{ptk} \geq 0, \qquad \forall p \in P,\ \forall t, k \in T,\ k \geq t,$$

$$w_{pt} \geq 0, \qquad \forall p \in P,\ \forall t \in T,$$

$$y_{pt} \in \{0, 1\}, \qquad \forall p \in P,\ \forall t \in T.$$

- This MIP model is referred to as the $\mathrm{MIP}^{\mathrm{SP,SI}}_{\mathrm{CLSP}}$ model.
- Relaxing constraints $y_{pt} \in \{0, 1\}$ into $0 \leq y_{pt} \leq 1$ and removing the integrality constraints of the quantities gives the LP model referred to as the $\mathrm{LP}^{\mathrm{SP,SI}}_{\mathrm{CLSP}}$ model.

Description of the model $\mathrm{MIP}_{\mathrm{CLSP}}^{\mathrm{SP,SI}}$

- The linear objective function minimizes the cost of the setups, of the initial inventory and of the arcs of the plans of all the products;
- constraints $\sum_k (w_{pk} + zv_{p1k}) = 1$ and $w_{p,t-1} + \sum_u zv_{p,u,t-1} - \sum_k zv_{ptk} = 0$ are the flow constraints of the network of every product, with the arcs w_{pt} of the initial inventory leaving node 0 ($|P|\, m$ linear constraints);
- constraints $\sum_k zv_{ptk} \leq y_{pt}$ force the setup of product p in period t if it is produced ($|P|\, m$ linear constraints), and constraints $\sum_p st_{pt}\, y_{pt} + \sum_p \sum_k vt_{pt}\, sd_{ptk}\, zv_{ptk} \leq cap_t$ impose the capacities, setup times included (m linear constraints);
- the integrality constraints keep integer the quantities and the initial inventories;
- finally, the domain constraints define the variables of the MIP model.

Integer quantities

- The quantity produced is $x_{pt} = \sum_{k=t}^{m} sd_{ptk}\, zv_{ptk}$, and the initial inventory is $si_p = \sum_{t \in T} sd_{p1t}\, w_{pt}$: the integrality constraints of the quantities keep them integer, as in the $\mathrm{ILP}^{\mathrm{SI}}_{\mathrm{CLSP}}$ model.
- With one product and $vt_t = 1$ these constraints are not needed: once the setups are fixed, what remains is a minimum cost flow problem, which has an integer optimal solution.
- With several products and production times $vt_{pt} > 1$, a fractional quantity can use the capacity better. Without the integrality constraints of the quantities the model is only a relaxation of the problem, and its optimal value can be smaller than $z(\mathrm{ILP}^{\mathrm{SI}}_{\mathrm{CLSP}})$ (next example).

Example: lower bounds with two products (1/2)

- We consider the instance of the $\mathrm{ILP}^{\mathrm{SI}}_{\mathrm{CLSP}}$ example.
- Ignoring the setups (in the costs and in the capacities), the optimal value is 575400, attained by the integer solution

$$x_{11} = 120, \quad x_{12} = 160, \quad x_{13} = 10,$$
$$x_{21} = 10, \quad x_{22} = 100, \quad x_{23} = 150, \quad s_{11} = 40, \quad si_1 = 20.$$

Example: lower bounds with two products (2/2)

The lower bounds and the optimal value of the formulations are:

Model	Value
Without setups	575 400.00
$\mathrm{LP}^{\mathrm{SI}}_{\mathrm{CLSP}}$ with $M_{pt} = sd_{ptm}$	638 882.20
$\mathrm{LP}^{\mathrm{SI}}_{\mathrm{CLSP}}$ with the strong M_{pt}	651 267.37
$\mathrm{LP}^{\mathrm{SP,SI}}_{\mathrm{CLSP}}$	668 928.57
$\mathrm{MIP}^{\mathrm{SP,SI}}_{\mathrm{CLSP}}$ without integrality constraints	676 450.00
$\mathrm{ILP}^{\mathrm{SI}}_{\mathrm{CLSP}}$ and $\mathrm{MIP}^{\mathrm{SP,SI}}_{\mathrm{CLSP}}$ (integer optimum)	677 400

Without the integrality constraints of the quantities the shortest-path model reaches 676 450 with the same setups and the fractional quantities $x_{11} = 59.5$ and $si_1 = 40.5$: period 1 then uses its whole capacity, $100 + 20 \cdot 59.5 + 110 + 10 \cdot 110 = 2500$.

Overtime: data and variables

- The data are those of the lot sizing problem with several products and setup times, but the time of each period is not a fixed capacity: in period t the working time in normal hours costs cw_t per unit, up to cap_t^N, and the overtime costs co_t per unit, up to cap_t^O.

- We introduce the integer variables x_{pt}, s_{pt} and the binary variables y_{pt} of the $\mathrm{ILP}^P_{\mathrm{CLSP}}$ model, and, for every period $t \in T$, the following **integer variables**:

$$\mu_t \in \mathbb{Z}_{\geq 0} \text{ (working time in normal hours)},$$
$$\nu_t \in \mathbb{Z}_{\geq 0} \text{ (overtime)}.$$

  Since the times vt_{pt} and st_{pt} are integer, the working times can be taken integer.

The model $\mathrm{ILP}^{\mathrm{OT}}_{\mathrm{CLSP}}$: objective and constraints

Using these variables, an Integer Linear Programming (ILP) model for the lot sizing problem with overtime (LS-OT) reads as follows:

$$\min \quad \sum_{p\in P}\sum_{t\in T}(sc_{pt}\,y_{pt} + vc_{pt}\,x_{pt} + hc_{pt}\,s_{pt}) + \sum_{t\in T}(cw_t\,\mu_t + co_t\,\nu_t)$$

$$\text{subject to} \quad s_{p,t-1} + x_{pt} = d_{pt} + s_{pt}, \quad \forall p \in P,\ \forall t \in T,$$

$$\sum_{p\in P}(st_{pt}\,y_{pt} + vt_{pt}\,x_{pt}) \le \mu_t + \nu_t, \qquad \forall t \in T,$$

$$\mu_t \le cap^N_t, \qquad \forall t \in T,$$

$$\nu_t \le cap^O_t, \qquad \forall t \in T,$$

$$x_{pt} \le M_{pt}\,y_{pt}, \quad \forall p \in P,\ \forall t \in T,$$

$$s_{p0} = s_{pm} = 0, \qquad \forall p \in P,$$

The model $\mathrm{ILP}^{\mathrm{OT}}_{\mathrm{CLSP}}$: domains of the variables

$$
\begin{aligned}
x_{pt} &\in \mathbb{Z}_{\geq 0}, && \forall p \in P,\ \forall t \in T,\\
s_{pt} &\in \mathbb{Z}_{\geq 0}, && \forall p \in P,\ \forall t \in T,\\
\mu_t &\in \mathbb{Z}_{\geq 0}, && \forall t \in T,\\
\nu_t &\in \mathbb{Z}_{\geq 0}, && \forall t \in T,\\
y_{pt} &\in \{0,1\}, && \forall p \in P,\ \forall t \in T.
\end{aligned}
$$

This ILP model is referred to as the $\mathrm{ILP}^{\mathrm{OT}}_{\mathrm{CLSP}}$ model.

Description of the model $\mathrm{ILP}^{\mathrm{OT}}_{\mathrm{CLSP}}$

- The linear objective function adds to the costs of the $\mathrm{ILP}^{P}_{\mathrm{CLSP}}$ model the cost of the working time in normal hours and in overtime;
- constraints $s_{p,t-1} + x_{pt} = d_{pt} + s_{pt}$ are the flow balancing constraints of every product ($|P|\, m$ linear constraints);
- constraints $\sum_p (st_{pt}\, y_{pt} + vt_{pt}\, x_{pt}) \leq \mu_t + \nu_t$ impose that the setup times and the production times of each period are covered by the working time, in normal hours and in overtime (m linear constraints);
- constraints $\mu_t \leq cap^N_t$ and $\nu_t \leq cap^O_t$ bound the two working times (m linear constraints each);
- constraints $x_{pt} \leq M_{pt}\, y_{pt}$ link the quantities with the setups ($|P|\, m$ linear constraints);
- constraints $s_{p0} = s_{pm} = 0$ impose an empty inventory at the beginning and at the end of the horizon;
- finally, the domain constraints define the variables of the ILP model.

Strong big-M values with overtime

- The time available in period t is at most $cap_t^N + cap_t^O$, so the strong big-M value is

$$M_{pt} = \min \left\{ \left\lfloor \frac{cap_t^N + cap_t^O - st_{pt}}{vt_{pt}} \right\rfloor, \; sd_{ptm} \right\}.$$

- With $co_t > cw_t$ an optimal solution uses overtime only when the normal hours of the period are exhausted.

Example: the $\mathrm{ILP}^{\mathrm{OT}}_{\mathrm{CLSP}}$ model (1/5)

- We consider the CLS reference instance, with one product ($|P| = 1$, and we drop the index p), production times $vt_t = 1$ and setup times $st_t = 0$.
- The capacity $cap_t = 4$ becomes the normal hours $cap^N_t = 4$, at the cost $cw_t = 1$, and one unit of overtime is available in every period: $cap^O_t = 1$, at the cost $co_t = 6$.
- The big-M values are $M_t = \min\{5, D_{t,5}\}$, that is $(5, 5, 5, 5, 1)$.

The $\mathrm{ILP}^{\mathrm{OT}}_{\mathrm{CLSP}}$ model for this instance is:

$$\begin{aligned}\min \quad & 50\,y_1 + 50\,y_2 + 50\,y_3 + 50\,y_4 + 50\,y_5 + 10\,s_1 \\ & +10\,s_2 + 10\,s_3 + 10\,s_4 + 10\,s_5 + \mu_1 + \mu_2 \\ & +\mu_3 + \mu_4 + \mu_5 + 6\,\nu_1 + 6\,\nu_2 + 6\,\nu_3 \\ & +6\,\nu_4 + 6\,\nu_5\end{aligned}$$

Example: the $\mathrm{ILP}^{\mathrm{OT}}_{\mathrm{CLSP}}$ model (2/5)

$$
\begin{aligned}
\text{subject to} \quad & x_1 - s_1 && = && 2 \\
& s_1 + x_2 - s_2 && = && 1 \\
& s_2 + x_3 - s_3 && = && 3 \\
& s_3 + x_4 - s_4 && = && 4 \\
& s_4 + x_5 - s_5 && = && 1 \\
& x_1 - \mu_1 - \nu_1 && \leq && 0 \\
& x_2 - \mu_2 - \nu_2 && \leq && 0 \\
& x_3 - \mu_3 - \nu_3 && \leq && 0 \\
& x_4 - \mu_4 - \nu_4 && \leq && 0 \\
& x_5 - \mu_5 - \nu_5 && \leq && 0
\end{aligned}
$$

Example: the $\mathrm{ILP}^{\mathrm{OT}}_{\mathrm{CLSP}}$ model (3/5)

$$
\begin{aligned}
x_1 - 5\,y_1 &\leq 0 \\
x_2 - 5\,y_2 &\leq 0 \\
x_3 - 5\,y_3 &\leq 0 \\
x_4 - 5\,y_4 &\leq 0 \\
x_5 - y_5 &\leq 0 \\
\mu_1 &\leq 4 \\
\mu_2 &\leq 4 \\
\mu_3 &\leq 4 \\
\mu_4 &\leq 4 \\
\mu_5 &\leq 4
\end{aligned}
$$

Example: the $\mathrm{ILP}^{\mathrm{OT}}_{\mathrm{CLSP}}$ model (4/5)

$$
\begin{aligned}
\nu_1 &\leq 1 \\
\nu_2 &\leq 1 \\
\nu_3 &\leq 1 \\
\nu_4 &\leq 1 \\
\nu_5 &\leq 1 \\
s_5 &= 0 \\
\\
x_1,\ x_2,\ x_3,\ x_4,\ x_5 &\in \mathbb{Z}_{\geq 0} \\
s_1,\ s_2,\ s_3,\ s_4,\ s_5 &\in \mathbb{Z}_{\geq 0} \\
\mu_1,\ \mu_2,\ \mu_3,\ \ldots,\ \nu_5 &\in \mathbb{Z}_{\geq 0} \\
y_1,\ y_2,\ y_3,\ y_4,\ y_5 &\in \{0,1\}
\end{aligned}
$$

Example: the $\mathrm{ILP}^{\mathrm{OT}}_{\mathrm{CLSP}}$ model (5/5)

An optimal solution of the $\mathrm{ILP}^{\mathrm{OT}}_{\mathrm{CLSP}}$ model is:

$$\tilde{\boldsymbol{y}} = (1,0,1,1,0), \qquad \tilde{\boldsymbol{x}} = (3,0,3,5,0), \qquad \tilde{\boldsymbol{s}} = (1,0,0,1,0),$$
$$\tilde{\boldsymbol{\mu}} = (3,0,3,4,0), \qquad \tilde{\boldsymbol{\nu}} = (0,0,0,1,0)$$

- The optimal value $z(\mathrm{ILP}^{\mathrm{OT}}_{\mathrm{CLSP}})$ is $3 \cdot 50 + 2 \cdot 10 + 10 \cdot 1 + 1 \cdot 6 = 186$, and the optimal solution is unique (Gurobi).
- One unit of overtime in period 4, at the cost 6, replaces one unit stored from period 3 to period 4, at the cost 10: the optimal plan $(3,0,4,4,0)$ of the CLS costs 191 here.

Unsatisfied demand: data and variables

- The data are those of the lot sizing problem with several products and setup times. In addition, a part of the demand d_{pt} may be left unsatisfied, at the penalty cu_{pt} per unit.

- We introduce the integer variables x_{pt}, s_{pt} and the binary variables y_{pt} of the $\mathrm{ILP}^{P}_{\mathrm{CLSP}}$ model, and, for every product $p \in P$ and every period $t \in T$, the following **integer variable**:

  $\varepsilon_{pt} \in \mathbb{Z}_{\geq 0}$ (demand of product p in period t left unsatisfied).

The model $\mathrm{ILP}^{\mathrm{UD}}_{\mathrm{CLSP}}$: objective and constraints

Using these variables, an Integer Linear Programming (ILP) model for the lot sizing problem with unsatisfied demand (LS-UD) reads as follows:

$$\min \quad \sum_{p\in P}\sum_{t\in T}(sc_{pt}\, y_{pt} + vc_{pt}\, x_{pt} + hc_{pt}\, s_{pt}) + \sum_{p\in P}\sum_{t\in T} cu_{pt}\, \varepsilon_{pt}$$

$$\text{subject to} \quad s_{p,t-1} + x_{pt} = d_{pt} - \varepsilon_{pt} + s_{pt}, \quad \forall p \in P,\ \forall t \in T,$$

$$\sum_{p\in P}(st_{pt}\, y_{pt} + vt_{pt}\, x_{pt}) \le cap_t, \qquad \forall t \in T,$$

$$x_{pt} \le M_{pt}\, y_{pt}, \qquad \forall p \in P,\ \forall t \in T,$$

$$\varepsilon_{pt} \le d_{pt}, \qquad \forall p \in P,\ \forall t \in T,$$

$$s_{p0} = s_{pm} = 0, \qquad \forall p \in P,$$

The model $\mathrm{ILP}^{\mathrm{UD}}_{\mathrm{CLSP}}$: domains of the variables

$$
\begin{aligned}
x_{pt} &\in \mathbb{Z}_{\geq 0}, && \forall p \in P,\ \forall t \in T,\\
s_{pt} &\in \mathbb{Z}_{\geq 0}, && \forall p \in P,\ \forall t \in T,\\
\varepsilon_{pt} &\in \mathbb{Z}_{\geq 0}, && \forall p \in P,\ \forall t \in T,\\
y_{pt} &\in \{0,1\}, && \forall p \in P,\ \forall t \in T.
\end{aligned}
$$

This ILP model is referred to as the $\mathrm{ILP}^{\mathrm{UD}}_{\mathrm{CLSP}}$ model.

Description of the model $\mathrm{ILP}^{\mathrm{UD}}_{\mathrm{CLSP}}$

- The linear objective function adds to the costs of the $\mathrm{ILP}^{P}_{\mathrm{CLSP}}$ model the penalty of the unsatisfied demand;
- constraints $s_{p,t-1} + x_{pt} = d_{pt} - \varepsilon_{pt} + s_{pt}$ are the flow balancing constraints, with the demand reduced by its unsatisfied part ($|P|\, m$ linear constraints);
- constraints $\sum_p (st_{pt}\, y_{pt} + vt_{pt}\, x_{pt}) \leq cap_t$ are the capacity constraints and constraints $x_{pt} \leq M_{pt}\, y_{pt}$ link the quantities with the setups (m and $|P|\, m$ linear constraints);
- constraints $\varepsilon_{pt} \leq d_{pt}$ ensure that the unsatisfied demand does not exceed the demand ($|P|\, m$ linear constraints);
- constraints $s_{p0} = s_{pm} = 0$ impose an empty inventory at the beginning and at the end of the horizon;
- finally, the domain constraints define the variables of the ILP model.

Big-M values with unsatisfied demand

- The big-M values of the $\mathrm{ILP}^{P}_{\mathrm{CLSP}}$ model remain valid, since the production can only decrease.
- With $\varepsilon_{pt} = 0$ for every product and period the model is the $\mathrm{ILP}^{P}_{\mathrm{CLSP}}$ model.

Example: the $\mathrm{ILP}^{\mathrm{UD}}_{\mathrm{CLSP}}$ model (1/4)

We consider the CLS reference instance, with one product ($|P| = 1$, and we drop the index p), production times $vt_t = 1$, setup times $st_t = 0$ and the strong big-M values $M_t = (4, 4, 4, 4, 1)$. Each unit of unsatisfied demand costs $cu_t = 18$. The $\mathrm{ILP}^{\mathrm{UD}}_{\mathrm{CLSP}}$ model for this instance is:

$$\begin{aligned} \min \quad & 50\,y_1 + 50\,y_2 + 50\,y_3 + 50\,y_4 + 50\,y_5 + 10\,s_1 \\ & +10\,s_2 + 10\,s_3 + 10\,s_4 + 10\,s_5 + 18\,\varepsilon_1 + 18\,\varepsilon_2 \\ & +18\,\varepsilon_3 + 18\,\varepsilon_4 + 18\,\varepsilon_5 \end{aligned}$$

Example: the $\mathrm{ILP}^{\mathrm{UD}}_{\mathrm{CLSP}}$ model (2/4)

$$
\begin{array}{lrcl}
\text{subject to} & x_1 - s_1 + \varepsilon_1 & = & 2 \\
& s_1 + x_2 - s_2 + \varepsilon_2 & = & 1 \\
& s_2 + x_3 - s_3 + \varepsilon_3 & = & 3 \\
& s_3 + x_4 - s_4 + \varepsilon_4 & = & 4 \\
& s_4 + x_5 - s_5 + \varepsilon_5 & = & 1 \\
& x_1 & \leq & 4 \\
& x_2 & \leq & 4 \\
& x_3 & \leq & 4 \\
& x_4 & \leq & 4 \\
& x_5 & \leq & 4
\end{array}
$$

Example: the $\mathrm{ILP}^{\mathrm{UD}}_{\mathrm{CLSP}}$ model (3/4)

$$
\begin{aligned}
x_1 - 4\,y_1 &\leq 0 \\
x_2 - 4\,y_2 &\leq 0 \\
x_3 - 4\,y_3 &\leq 0 \\
x_4 - 4\,y_4 &\leq 0 \\
x_5 - y_5 &\leq 0 \\
\varepsilon_1 &\leq 2 \\
\varepsilon_2 &\leq 1 \\
\varepsilon_3 &\leq 3 \\
\varepsilon_4 &\leq 4 \\
\varepsilon_5 &\leq 1
\end{aligned}
$$

Example: the $\mathrm{ILP}^{\mathrm{UD}}_{\mathrm{CLSP}}$ model (4/4)

$$
\begin{array}{lcl}
s_5 & = & 0 \\
x_1,\ x_2,\ x_3,\ x_4,\ x_5 & \in & \mathbb{Z}_{\geq 0} \\
s_1,\ s_2,\ s_3,\ s_4,\ s_5 & \in & \mathbb{Z}_{\geq 0} \\
\varepsilon_1,\ \varepsilon_2,\ \varepsilon_3,\ \varepsilon_4,\ \varepsilon_5 & \in & \mathbb{Z}_{\geq 0} \\
y_1,\ y_2,\ y_3,\ y_4,\ y_5 & \in & \{0,1\}
\end{array}
$$

An optimal solution of the $\mathrm{ILP}^{\mathrm{UD}}_{\mathrm{CLSP}}$ model is:

$$\tilde{\boldsymbol{y}} = (0,0,1,1,0), \qquad \tilde{\boldsymbol{x}} = (0,0,3,4,0),$$

$$\tilde{\boldsymbol{s}} = (0,0,0,0,0), \qquad \tilde{\boldsymbol{\varepsilon}} = (2,1,0,0,1)$$

Example: the $\mathrm{ILP}^{\mathrm{UD}}_{\mathrm{CLSP}}$ model, the optimal value

- The optimal value $z(\mathrm{ILP}^{\mathrm{UD}}_{\mathrm{CLSP}})$ is $2 \cdot 50 + 4 \cdot 18 = 172$, and the optimal solution is unique (Gurobi).
- The demands of the periods 1, 2 and 5 are not satisfied: their 4 units cost 72, and the plan saves one setup and all the holding costs of the optimal plan $(3, 0, 4, 4, 0)$ of the CLS, of cost 180.

Summary (1/2)

The uncapacitated lot sizing problem:

- The big-M model without capacities and with $M_t = D_{t,m}$
- On the reference instance its optimal value is 170, the bound of the shortest-path formulation of the CLS

Several products and setup times:

- The products share the capacity with their setup times; they interact only through the capacity constraints
- Strong big-M: $M_{pt} = \min\{\lfloor (cap_t - st_{pt})/vt_{pt} \rfloor, sd_{ptm}\}$

Summary (2/2)

Initial stock: the variables si_p, at the cost fc_p; with several products the shortest-path model needs the integrality of the quantities ($676\,450 < 677\,400$)

Overtime: the working times μ_t and ν_t replace the fixed capacity; the big-M values use $cap_t^N + cap_t^O$; on the reference instance the optimal value is 186

Unsatisfied demand: the variables $\varepsilon_{pt} \leq d_{pt}$, with the penalty cu_{pt}; on the reference instance the optimal value is 172

Typical mistakes

Introduction: typical mistakes

- We collect the mistakes that are most frequent when writing the lot sizing models
- They concern the big-M values, the capacity constraints, the costs of the arcs and the variants
- Each one comes with the reason why it is a mistake

Observation 1: Typical mistakes (1/3)

- ***Taking a big-M value that is too small.*** *A value smaller than the production of a feasible plan cuts off that plan: the values $D_{t,m}$ and cap_t are both valid.*
- ***Taking a big-M value that is too large.*** *Every valid value gives a correct model, but a large value gives a weak relaxation: the setup variables become almost zero in the LP solutions, as in the bound 99.65 of the reference instance.*
- ***Forgetting the setup times in the capacity constraints.*** *With several products the constraints contain $st_{pt}\, y_{pt}$ besides $vt_{pt}\, x_{pt}$.*

Observation: Typical mistakes (2/3)

- ***Counting the holding cost of the demand of the production period.*** *In cv_{tk} the units for period t are not stored: the holding sum starts from the demand of period $t+1$.*
- ***Believing that the shortest-path formulation needs binary paths.*** *The variables zv_{tk} are continuous: a feasible plan that is not a path is a combination of paths.*
- ***Forgetting the integrality of the quantities with several products.*** *With production times $vt_{pt} > 1$ the continuous variables zv_{ptk} can give fractional quantities: the shortest-path model needs the integrality constraints of the quantities, otherwise it is only a relaxation ($676\,450 < 677\,400$).*

Observation: Typical mistakes (3/3)

- ***Keeping the fixed capacity in the model with overtime.*** *The time of period t is bounded by the working time $\mu_t + \nu_t$, and the big-M values use $cap_t^N + cap_t^O$.*
- ***Leaving the unsatisfied demand unbounded.*** *Without $\varepsilon_{pt} \leq d_{pt}$ the balance constraints would allow a negative net demand, that is, inventory created from nothing at the price of the penalty.*

Summary

Big-M values:

- Too small cuts off feasible plans; too large gives a weak relaxation

Shortest-path formulation:

- The holding sum of cv_{tk} starts from period $t + 1$; the variables zv_{tk} are continuous
- With several products the integrality of the quantities is needed

Variants:

- Setup times in the capacity constraints; working time $\mu_t + \nu_t$ with overtime; $\varepsilon_{pt} \leq d_{pt}$

In summary

Introduction: in summary

- We collect the key ideas of the chapter
- We close with the dashboard of the lower bounds computed on the reference instance

Summary (1/3)

- Without the setups the CLS is a simple flow problem; the setups need the binary variables y_t and a link with the quantities.
- In the big-M formulation the link $x_t \leq M_t\, y_t$ is correct for every valid M_t, but the LP bound improves when M_t decreases: the strong value is $\min\{cap_t, D_{t,m}\}$.
- The shortest-path formulation writes the plans as paths in a network, with the link $\sum_k zv_{tk} \leq y_t$ and no big-M value: its relaxation is the strongest.

Summary (2/3)

- The two formulations have the same optimal value; they differ in the lower bound and in the size of the model.
- The ULS model is the big-M model without capacities and with $M_t = D_{t,m}$: on the reference instance its optimal value is 170.
- The variants keep the structure of the big-M model: several products share the capacity with their setup times, the initial stock adds the variables si_p, overtime adds the working times μ_t, ν_t, and the unsatisfied demand adds the variables ε_{pt}. On the reference instance, overtime gives 186 and unsatisfied demand 172.

Summary (3/3)

Dashboard of the reference instance. Lower bounds on the optimal value computed in this chapter:

Bound	Value	From
Without setups	0	lot-for-lot plan
Big-M, $M_t = D_{t,m}$	99.65	$z(\mathrm{LP}^{\mathrm{dem}}_{\mathrm{CLSP}}) = \frac{9865}{99}$
Big-M, $M_t = cap_t$	137.5	$z(\mathrm{LP}^{\mathrm{cap}}_{\mathrm{CLSP}})$
Strong big-M	157.5	$z(\mathrm{LP}^{\mathrm{strong}}_{\mathrm{CLSP}})$
Shortest path	170	$z(\mathrm{LP}^{\mathrm{SP}}_{\mathrm{CLSP}})$
Optimal value	180	plan $(3, 0, 4, 4, 0)$